

Find a Gene Assignment

Hubert Phan (PID: A17663330)

Q1. Tell me the name of a protein you are interested in. Include the species, accession number and known function. This can be a human protein or a protein from any other species as long as it's function is known.

If you do not have a favorite protein, select human RBP4 or KIF11. Do not use beta globin as this is in the worked example report that I provide you with online.

Name: Interleukin-36 gamma (IL36G)

Species: Homo sapiens

Accession: Q9NZH8

Function: Interleukin-36 gamma (IL36G) plays a role in skin inflammation and immune signaling. IL36G is associated in skin disorders such as psoriasis and dandruff.

Q2. Perform a BLAST search against a DNA database, such as a database consisting of genomic DNA or ESTs. The BLAST server can be at NCBI or elsewhere. Include details of the BLAST method used, database searched and any limits applied (e.g. Organism).

Method: TBLASTN search against reptilia nt

Database: Nucleotide collection (nt)

Organsim: Reptilia (taxid:8459)

Also include the output of that BLAST search in your document. If appropriate, change the font to Courierdd size 10 so that the results are displayed neatly. You can also screen capture a BLAST output (e.g. alt print screen on a PC or on a MAC press -shift-4. The pointer becomes a bulls eye. Select the area you wish to capture and release. The image is saved as a file called Screen Shot [].png in your Desktop directory). It is not necessary to print out all of the blast results if there are many pages.

National Library of Medicine
National Center for Biotechnology Information

hubertphan04@gmail...

BLAST® » tblastn
Home Recent Results Saved Strategies Help

Translated BLAST: tblastn

Use the core_nt database for faster and better results. Submit with core_nt

blastn blastp blastx **tblastn** tblastx

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) Clear

>sp|Q9NZH8|IL36G_HUMAN Interleukin-36 gamma OS=Homo sapiens
OX=9606 GN=IL36G PE=1 SV=1
MRGTPGDADGGGRAVYQSMCKPITGTINDLNQVWTLGGQNLVAPRSDSVT
PVTVAITCKYPEALEGG
RGDPYLGIONPEMCLYCEKVGEPQLQLKEQKIMDLYGQPEPVKPLFYRAKT
GRSTLTLESVAFPDWFI
ASSKRQGPRLTSELGKSYNTAFELNIND

Query subrange
From
To

Or, upload file No file chosen

Job Title
Enter a descriptive title for your BLAST search

☐ Align two or more sequences

Choose Search Set

Database

Organism Optional ☐ exclude
Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown

Exclude Optional ☐ Models (XM/XP) ☐ Uncultured/environmental sample sequences

Limit to Optional ☐ Sequences from type material

Entrez® Query Optional
Enter an Entrez query to limit search

BLAST

Search database nt using Tblastn (search translated nucleotide databases using a protein query)
☐ Show results in a new window

+ Algorithm parameters

Figure 1: TBLASTN QUERY

On the BLAST results, clearly indicate a match that represents a protein sequence, encoded from some DNA sequence, that is homologous to your query protein. I need to be able to inspect the pairwise alignment you have selected, including the E value and score. It should be labeled a “genomic clone” or “mRNA sequence”, etc. - but include no functional annotation.

Chosen Match: Accession OZ346534.1, *Trionyx triunguis* genome assembly. See alignment details below.

[Edit Search](#)
[Save Search](#)
[Search Summary](#)

[How to read this report?](#)
[BLAST Help Videos](#)
[Back to Traditional Results Page](#)

i Your search is limited to records include: **Reptilia (taxid:8459)**

Job Title	sp Q9NZH8 IL36G_HUMAN Interleukin-36 gamma...
RID	RU2HMSBS016 Search expires on 02-01 02:44 am Download All
Program	TBLASTN Citation
Database	nt See details
Query ID	lcl Query_6676515
Description	sp Q9NZH8 IL36G_HUMAN Interleukin-36 gamma OS=Hor...
Molecule type	amino acid
Query Length	169
Other reports	?

Filter Results

Organism only top 20 will appear ☐ exclude

Type common name, binomial, taxid or group name

[+ Add organism](#)

Percent Identity to E value to Query Coverage to

[Filter](#) [Reset](#)

Descriptions [Graphic Summary](#) [Alignments](#) [Taxonomy](#)

Sequences producing significant alignments [Download](#) [Select columns](#) [Show](#) [?](#)

☒ select all 73 sequences selected [GenBank](#) [Graphics](#)

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	PREDICTED: Chelonoidis abingdonii interleukin-36 receptor antagonist protein-like (LOC116819404), m...	Chelonoidis abi...	95.5	95.5	90%	1e-20	35.29%	3011	XM_032771100.2
☒	PREDICTED: Emys orbicularis interleukin-36 receptor antagonist protein-like (LOC135873734), mRNA	Emys orbicularis	87.8	87.8	89%	1e-19	35.53%	576	XM_065398370.1
☒	PREDICTED: Chelonoidis abingdonii interleukin-36 receptor antagonist protein-like (LOC116819405), m...	Chelonoidis abi...	87.8	87.8	89%	1e-19	34.87%	576	XM_032771101.1
☒	PREDICTED: Gopherus flavomarginatus interleukin-36 receptor antagonist protein-like (LOC127044660)...	Gopherus flavo...	87.4	87.4	89%	2e-19	34.87%	576	XM_050939738.1
☒	PREDICTED: Malaclemys terrapin pileata interleukin-36 receptor antagonist protein-like (LOC12883275...	Malaclemys terr...	90.1	90.1	85%	2e-19	36.81%	1090	XM_054020339.1
☒	PREDICTED: Mauremys reevesii interleukin-36 receptor antagonist protein-like (LOC120398387), transc...	Mauremys reev...	88.2	88.2	84%	3e-19	37.06%	764	XM_039525790.1
☒	PREDICTED: Mauremys mutica interleukin-36 receptor antagonist protein-like (LOC123364511), transcri...	Mauremys mutica	90.9	90.9	90%	3e-19	35.95%	2253	XM_045006715.1
☒	PREDICTED: Mauremys mutica interleukin-36 receptor antagonist protein-like (LOC123364511), transcri...	Mauremys mutica	90.9	90.9	90%	3e-19	35.95%	2267	XM_045006714.1
☒	PREDICTED: Mauremys mutica interleukin-36 receptor antagonist protein-like (LOC123364511), transcri...	Mauremys mutica	90.9	90.9	90%	3e-19	35.95%	2207	XM_045006713.1
☒	PREDICTED: Chrysemys picta bellii interleukin-36 receptor antagonist protein (LOC101949597), mRNA	Chrysemys pict...	90.1	180	85%	4e-19	36.81%	1357	XM_005279282.4
☒	PREDICTED: Trachemys scripta elegans interleukin-36 receptor antagonist protein-like (LOC117871587...	Trachemys scri...	90.1	90.1	85%	5e-19	36.81%	1389	XM_034759399.1
☒	PREDICTED: Mauremys reevesii interleukin-36 receptor antagonist protein-like (LOC120398387), transc...	Mauremys reev...	88.2	88.2	84%	5e-19	37.06%	836	XM_039525788.1
☒	PREDICTED: Emydura macquarii macquarii interleukin-36 receptor antagonist protein-like (LOC1371479...	Emydura macq...	87.8	87.8	96%	6e-19	32.75%	821	XM_067540356.1
☒	PREDICTED: Mauremys reevesii interleukin-36 receptor antagonist protein-like (LOC120398387), transc...	Mauremys reev...	88.2	88.2	84%	6e-19	37.06%	896	XM_039525787.1
☒	PREDICTED: Carettochelys insculpta interleukin-36 receptor antagonist protein-like (LOC142004348), m...	Carettochelys in...	85.5	85.5	91%	7e-19	34.19%	529	XM_074981942.1
☒	PREDICTED: Mauremys mutica interleukin-36 receptor antagonist protein-like (LOC123365076), transcri...	Mauremys mutica	88.2	88.2	82%	1e-18	36.69%	1072	XM_045007705.1
☒	PREDICTED: Emydura macquarii macquarii interleukin-36 receptor antagonist protein-like (LOC1371478...	Emydura macq...	83.6	83.6	90%	2e-17	33.33%	787	XM_067540265.1
☒	PREDICTED: Trachemys scripta elegans interleukin-36 receptor antagonist protein-like (LOC117871587...	Trachemys scri...	82.4	82.4	85%	2e-16	34.84%	1434	XM_034759398.1
☒	PREDICTED: Pelodiscus sinensis interleukin-1 family member 10-like (LOC142820928), transcript varia...	Pelodiscus sine...	79.0	79.0	82%	4e-15	35.25%	1692	XM_075910708.1
☒	PREDICTED: Pelodiscus sinensis interleukin-1 family member 10-like (LOC142820928), transcript varia...	Pelodiscus sine...	79.0	79.0	79%	5e-15	35.82%	1690	XM_075910709.1
☒	PREDICTED: Pelodiscus sinensis interleukin-1 family member 10-like (LOC142820928), transcript varia...	Pelodiscus sine...	78.6	78.6	79%	6e-15	35.82%	1857	XM_075910706.1
☒	PREDICTED: Mauremys mutica interleukin-36 receptor antagonist protein-like (LOC123365076), transcri...	Mauremys mutica	75.5	75.5	53%	4e-14	43.96%	949	XM_045007706.1
☒	PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC102448715), trans...	Pelodiscus sine...	75.1	75.1	90%	9e-14	31.37%	1468	XM_075910776.1
☒	PREDICTED: Carettochelys insculpta interleukin-36 receptor antagonist protein-like (LOC142004315), m...	Carettochelys in...	72.4	72.4	85%	2e-13	32.88%	653	XM_074981886.1
☒	PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC102448715), trans...	Pelodiscus sine...	73.6	73.6	90%	3e-13	31.37%	1428	XM_075910775.1
☒	PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC142820938), trans...	Pelodiscus sine...	71.6	71.6	88%	1e-12	31.13%	1290	XM_075910855.1
☒	PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC142820938), trans...	Pelodiscus sine...	71.2	71.2	88%	2e-12	31.13%	1290	XM_075910852.1
☒	PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC142820938), trans...	Pelodiscus sine...	71.2	71.2	88%	2e-12	31.13%	1180	XM_075910854.1
☒	PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC142820938), trans...	Pelodiscus sine...	70.9	70.9	88%	2e-12	31.13%	1283	XM_075910851.1

Figure 2: TBLASTN RESULTS

Trionyx triunguis genome assembly, chromosome: 30

Sequence ID: [OZ346534.1](#) Length: 17913073 Number of Matches: 1

Range 1: 1420032 to 1420238 [GenBank](#) [Graphics](#)

▼ [Next Match](#) ▲ [Previous Match](#)

Score	Expect	Method	Identities	Positives	Gaps	Frame	
50.1 bits(118)	5e-05	Compositional matrix adjust.	26/69(38%)	41/69(59%)	1/69(1%)	+3	
Query	91	VGEQPTLQLKEQKIMDLYGQPEVPKFLFYRAKTGRSTLESVAFPDWFIASSKRD-QPI					149
		+ P +++++IMDLY + K F F TG T ESVAFP WF+++S +D QP+					
Sbjct	1420032	LSHAPPPHPQDKEIMDLYQNSQEAKRFTFTCMATGSTRRFESVAFPGWFLSTSPKDGQPL					1420211
Query	150	ILTSELGKS					158
		+T+ G++					
Sbjct	1420212	RVTNISGEA					1420238

Figure 3: Alignment Details

In general, [Q2] is the most difficult for students because it requires you to have a “feel” for how to interpret BLAST results. You need to distinguish between a perfect match to your query (i.e. a sequence that is not “novel”), a near match (something that might be “novel”, depending on the results of [Q4]), and a non-homologous result.

If you are having trouble finding a novel gene try restricting your search to an organism that is poorly annotated.

Q3. Gather information about this “novel” protein. At a minimum, show me the protein sequence of the “novel” protein as displayed in your BLAST results from [Q2] as FASTA format (you can copy and paste the aligned sequence subject lines from your BLAST result page if necessary) or translate your novel DNA sequence using a tool called EMBOSS Transeq at the EBI. Don’t forget to translate all six reading frames; the ORF (open reading frame) is likely to be the longest sequence without a stop codon. It may not start with a methionine if you don’t have the complete coding region. Make sure the sequence you provide includes a header/subject line and is in traditional FASTA format.

The genomic DNA region surrounding the tblastn hit in *Trionyx triunguis* was extracted from the genome assembly and expanded beyond the aligned coordinates to capture potential coding sequence. The sequence was then translated in all six reading frames using EMBOSS Transeq. The longest open reading frame (ORF) without a stop codon was identified by examining for conserved motifs observed in the original BLAST alignment. This segment is the protein sequence used for downstream BLASTP analysis to assess sequence novelty.

Chosen Sequence:

```
>Trionyx_triunguis_tblastn_hit_translated_fragment_frame_2
GQWADGRSLPPSLQAVSPFLIGLSHAPPPHPQDKEIMDLYQNSQEAKRFTFTCMATGSTRRFESVAFPGW
FLSTSPKDGQPLRVTNISGEAEIINFYFKKM
```

Here, tell me the name of the novel protein, and the species from which it derives. It is very unlikely (but still definitely possible) that you will find a novel gene from an organism such as *S. cerevisiae*, human or mouse, because those genomes have already

been thoroughly annotated. It is more likely that you will discover a new gene in a genome that is currently being sequenced, such as bacteria or plants or protozoa.

Name: IL-1 family receptor antagonist-like protein

Species: *Trionyx triunguis* (African softshell turtle)

Lineage: Eukaryota; Metazoa; Chordata; Reptilia; Testudines; Trionychidae; *Trionyx*

Q4. For the purposes of this project, “novel” is defined as follows. Take the protein sequence (your answer to [Q3]) and use it as a query in a BLASTP search of the nr database at NCBI.

- If there is a match with **100% amino acid identity** to a protein in the database **from the same species**, then your protein is **NOT novel** (even if the match is to a protein with a name such as “unknown”). Someone has already found and annotated this sequence and assigned it an accession number.
- If the **top match reported has less than 100% identity**, then it is likely that your protein is **novel**, and you have succeeded.
- If there is a match with **100% identity**, but to a **different species** than the one you started with, then you have likely succeeded in finding a novel gene.
- If there are **no database matches** to the original query from [Q1], this indicates that you have **partially succeeded**: yes, you may have found a new gene, but no, it is not actually homologous to the original query. You should probably start over.

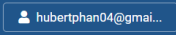
Details:

A BLASTP search against the nr database yielded a top hit protein from *Pelodiscus sinensis* (Chinese softshell turtle).

Below are screenshots of the BLASTP query, BLASTP results, and alignment details of the top hit.



National Library of Medicine
National Center for Biotechnology Information


hubertphan04@gmail...

BLAST® » blastp suite
Home
Recent Results
Saved Strategies
Help

blastn
blastp
blastx
tblastn
tblastx

Standard Protein BLAST

BLASTP programs search protein databases using a protein query, more...
Reset page
Bookmark

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s)
Clear

>Trionyx triunguis_blastn_hit translated fragment frame_2 (OZ346534.1
chr30:1419032..1421238)
GQWADGRSLPPSLQAVSPFLGLSHAPPPHPQDKIMDLYQNSQEAKRFTFT
CMATGSTRRFESWAFPGW

Query subrange
From
To

Or, upload file
Choose File
No file chosen

Job Title
Trionyx triunguis_blastn_hit_translated_fragment...
Enter a descriptive title for your BLAST search

☐ Align two or more sequences

Choose Search Set

Database
Non-redundant protein sequences (nr)

Organism
Optional
Enter organism name or id—completions will be suggested
exclude
Add organism
Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown

Exclude
Optional
☐ Models (XM/XP)
☐ Non-redundant RefSeq proteins (WP)
☐ Uncultured/environmental sample sequences

Program Selection

Algorithm
☐ Quick BLASTP (Accelerated protein-protein BLAST)
☒ blastp (protein-protein BLAST)
☐ PSI-BLAST (Position-Specific Iterated BLAST)
☐ PHI-BLAST (Pattern Hit Initiated BLAST)
☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)
Choose a BLAST algorithm

BLAST

Search database nr using Blastp (protein-protein BLAST)
☐ Show results in a new window

Note: Parameter values that differ from the default are highlighted in yellow and marked with + sign

+ Algorithm parameters

Figure 4: BLASTP Query

[< Edit Search](#)
[Save Search](#)
[Search Summary](#)

[How to read this report?](#)
[BLAST Help Videos](#)
[Back to Traditional Results Page](#)

Job Title Trionyx_triunguis_tblastn_hit_translated_fragment...
 RID [RV67KW6H014](#) Search expires on 02-01 12:53 pm [Download All](#)
Program BLASTP [Citation](#)
Database nr [See details](#)
Query ID lcl|Query_10407620
 Description Trionyx_triunguis_tblastn_hit_translated_fragment frame_2 ...
 Molecule type amino acid
 Query Length 101
 Other reports [Distance tree of results](#) [Multiple alignment](#) [MSA viewer](#)

Filter Results

Organism only top 20 will appear ☐ exclude

[+ Add organism](#)

Percent Identity to
E value to
Query Coverage to

[Filter](#)
[Reset](#)

Compare these results against the new Clustered nr database [?](#)

[BLAST](#)

Descriptions
Graphic Summary
Alignments
Taxonomy

Sequences producing significant alignments
[Download](#)
[Select columns](#)
Show

☒ select all 100 sequences selected

[GenPept](#)
[Graphics](#)
[Distance tree of results](#)
[Multiple alignment](#)
[MSA Viewer](#)

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	interleukin-1 family member 10-like isoform X4 [Pelodiscus sinensis]	Pelodiscus sinensis	141	141	79%	3e-39	85.00%	236	XP_075766824.1
☒	interleukin-1 family member 10-like isoform X3 [Pelodiscus sinensis]	Pelodiscus sinensis	142	142	80%	4e-39	83.95%	255	XP_075766823.1
☒	uncharacterized protein LOC142820928 isoform X2 [Pelodiscus sinensis]	Pelodiscus sinensis	142	142	80%	8e-39	83.95%	302	XP_075766822.1
☒	uncharacterized protein LOC142820928 isoform X1 [Pelodiscus sinensis]	Pelodiscus sinensis	142	142	69%	2e-38	92.86%	344	XP_075766821.1
☒	interleukin-36 receptor antagonist protein-like [Chelonoidis abingdonii]	Chelonoidis abingdonii	124	124	80%	6e-33	74.07%	191	XP_032626992.1
☒	interleukin-36 receptor antagonist protein-like [Emys orbicularis]	Emys orbicularis	124	124	79%	6e-33	75.00%	191	XP_065254442.1
☒	interleukin-36 receptor antagonist protein-like isoform X2 [Trachemys scripta elegans]	Trachemys scripta elegans	125	125	80%	1e-32	75.31%	275	XP_034615290.1
☒	interleukin-36 receptor antagonist protein-like [Malaclemys terrapin pileata]	Malaclemys terrapin pileata	125	125	80%	1e-32	75.31%	275	XP_053876314.1
☒	interleukin-36 receptor antagonist protein [Chrysemys picta bellii]	Chrysemys picta bellii	124	124	80%	2e-32	74.07%	254	XP_005279339.2
☒	interleukin-1 receptor antagonist protein-like isoform X1 [Trachemys scripta elegans]	Trachemys scripta elegans	125	125	80%	3e-32	75.31%	290	XP_034615289.1
☒	tetracycline resistance MFS efflux pump [Platysternon megacephalum]	Platysternon megacephalum	122	122	79%	1e-31	73.75%	236	TFJ97238.1
☒	interleukin-36 receptor antagonist protein-like [Gopherus flavomarginatus]	Gopherus flavomarginatus	120	120	80%	2e-31	72.84%	191	XP_050795695.1
☒	hypothetical protein KIL84_011473 [Mauremys mutica]	Mauremys mutica	121	121	79%	2e-31	72.50%	219	KAH1177771.1
☒	interleukin-36 receptor antagonist protein-like isoform X2 [Mauremys mutica]	Mauremys mutica	120	120	80%	2e-31	71.60%	200	XP_044863641.1
☒	interleukin-36 receptor antagonist protein-like isoform X1 [Mauremys mutica]	Mauremys mutica	120	120	80%	4e-31	71.60%	241	XP_044863640.1
☒	unnamed protein product [Caretta caretta]	Caretta caretta	112	112	69%	1e-28	74.29%	150	CAM4704126.1

Figure 5: BLASTP Results

[Download](#)
[GenPept](#)
[Graphics](#)
[Next](#)
[Previous](#)
[Descriptions](#)

interleukin-1 family member 10-like isoform X4 [Pelodiscus sinensis]
Sequence ID: [XP_075766824.1](#) Length: 236 Number of Matches: 1

Range 1: 157 to 236
[GenPept](#)
[Graphics](#)
[Next Match](#)
[Previous Match](#)

Score	Expect	Method	Identities	Positives	Gaps
141 bits(356)	3e-39	Compositional matrix adjust.	68/80(85%)	72/80(90%)	0/80(0%)

Query 22 GLSHAPPPHPQDKEIMDLYQNSQEAKRFTFTCMATGSTRRFESVAFPGWFLSTSPKDGQP 81

G S P +DKEIM LYQNSQEAKRFTFTC+ATGSTRRFESVAFPGWFLSTSPKDGQP

Sbjct 157 GTSAPPTLMLEDKEIMALYQNSQEAKRFTFTCVATGSTRRFESVAFPGWFLSTSPKDGQP 216

Query 82 LRVTNISGEAEIINFYFKKM 101

+RVTNISGEAEI+NFYFKKM

Sbjct 217 VRVTNISGEAEIVNFYFKKM 236

Related Information
[Genome Data Viewer](#) - aligned genomic context

[Download](#)
[GenPept](#)
[Graphics](#)
[Next](#)
[Previous](#)
[Descriptions](#)

interleukin-1 family member 10-like isoform X3 [Pelodiscus sinensis]
Sequence ID: [XP_075766823.1](#) Length: 255 Number of Matches: 1

Range 1: 175 to 255
[GenPept](#)
[Graphics](#)
[Next Match](#)
[Previous Match](#)

Score	Expect	Method	Identities	Positives	Gaps
142 bits(357)	4e-39	Compositional matrix adjust.	68/81(84%)	72/81(88%)	0/81(0%)

Query 21 IGLSHAPPPHPQDKEIMDLYQNSQEAKRFTFTCMATGSTRRFESVAFPGWFLSTSPKDGQ 80

G S P +DKEIM LYQNSQEAKRFTFTC+ATGSTRRFESVAFPGWFLSTSPKDGQ

Sbjct 175 CGTSAPPTLMLEDKEIMALYQNSQEAKRFTFTCVATGSTRRFESVAFPGWFLSTSPKDGQ 234

Query 81 PLRVTNISGEAEIINFYFKKM 101

P+RVTNISGEAEI+NFYFKKM

Sbjct 235 PVRVTNISGEAEIVNFYFKKM 255

Related Information
[Genome Data Viewer](#) - aligned genomic context

Figure 6: Alignment Details

Q5. Generate a multiple sequence alignment with your novel protein, your original query protein, and a group of other members of this family from different species. A typical number of proteins to use in a multiple sequence alignment for this assignment purpose is a minimum of 5 and a maximum of 20 - although the exact number is up to you. Include the multiple sequence alignment in your report. Use Courier font with a size appropriate to fit page width. Side-note: Indicate your sequence in the alignment by choosing an appropriate name for each sequence in the input unaligned sequence file (i.e. edit the sequence file so that the species, or short common, names (rather than accession numbers) display in the output alignment and in the subsequent answers below). The goal in this step is to create an interesting an alignment for building a phylogenetic tree that illustrates species divergence.

Re-labeled sequences for alignment:

```
>Trionyx_triunguis IL-1 family-like protein frame_2 (novel protein)
GQWADGRSLPPLQAVSPFLIGLSHAPPPHPQDKEIMDLYQNSQEAKRFTFTCMATGSTRRFESVAFPGW
FLSTSPKDGQPLRVTNISGEAEIINFYFKKM

>IL36G_HUMAN
MRGTPGDADGGGRAVYQSMCKPITGTINDLNQQVWTLQGQNLVAVPRSDSVTPVTVAVIT
CKYPEALEQGRGDIYLGIQNPEMCLYCEKVGEQPTLQLKEQKIMDLYGQPEPVKPFIFY
RAKTGRSTLESVAFPDWFIASSKRDQPIILTSELGKSYNTAFELNIND

>Pelodiscus_sinensis XP_075766824.1 interleukin-1 family member 10-like isoform X4
MVSRTDNGIRATSPLEGAAEEDSFPSQAPETSTDVGMGSSRPAPPQDTAGTEPWGAGPRPDSWAQPSN
RDLVELFDEFLNQKQFIPSPVKHSLFTLRDTNQVRIRLHKQLVATPQTSNAPPEKISVVPNQFLDPSN
FPIIMGIDNGTRCLSCGTSAPPTLMLEDKEIMALYQNSQEAKRFTFTCVATGSTRRFESVAFPGWFLSTS
PKDGQPV RVTNISGEAEIVNFYFKKM
```


>Chelonoidis_abingdonii XP_032626992.1 interleukin-36 receptor antagonist protein-like
MTSTGHTSPAWPAGARTPGPYSPQDNSNLVDLFNKFFSQRPAMITVPGPSLFTLRDTSQKVVRCHHNRLM
ASPQTANTLPEKISVVPNQFMDPSHFPIIMGIDGGTRCLSCGTSAQPTLMLEDKKIMDLYQNSQEAKRFT
FTCSATGSTHREFSVAFFPGWYLSTSPRNDQPLQVTNRLGEAEITNFYFKKV

>Emys_orbicularis XP_065254442.1 interleukin-36 receptor antagonist protein-like
MTGTGHTTPARPGAGRTPGSYSPQGNLNLVNEFFKQRPITMAVPGPSLFTLRDTSQKVVRYYHNNRLV
ASPQTANAPPEKISVVPNQFMDPSHFPIIMGINGGTRCLSCGTSAQPTLMLEDKKIMDLYQNSREAKRFT
FTCSATGSTLRFESVAFFPGWYLSTSPRNDQPLRVNRLGEAEITNFYFKKV

>Trachemys_scripta_elegans XP_034615290.1 interleukin-36 receptor antagonist protein-like isoform X2
MVSSLLALLLTCCSFDCFSQSMNTNGIATPLPGEDETEEDAFLSQVRKTSTDGLDSDNPEGPVSAA
GSASTHTPPCSHQDMGTGNMSPAQPGAGRTPGPYSPQGNLNLVNEFFKQGPITMAVPRPSLFTLRD
TSQKVVRCHHNHLVASPQTANAPPEKISVVPNQFMDPSHFPIIMGINGGTRCLSCGTSAQPTLMLEDKKI
MDLYQNSQEAKRFTFTCSATGSTLRFESVAFFPGWYLSTSPRNDQPLRVNRLGEAEITNFYFKKV

>Chelonia_mydas XP_043392104.1 interleukin-36 receptor antagonist protein
MSISDIPANAADAGVLAQANVPRDQEELRLSRDSCSVFPEKTSVVPNQFMDPSHFPIVMGIDGGLCCLSC
STSAQPTLMLEDKKIMDLYQNSWEAERFTFTCSASSSTLRFELVAFPGWYLSTSPRNDQPLWVITYHLGEA
EITNFYFKKV

>Podarcis_cretensis XP_079811393.1 interleukin-36 receptor antagonist protein-like isoform 1
MEKVDTETLREKPHKMTVDQKMRDLFHHFKPDHSHKFPPISLDRPWLYRIWDINQKFLFLKNNMLVAAPK
DGNSPDYLVAVTPNRGMDEKNSPIFLGTQDGAQTLSCGESGGQPQLTLEHKAIMDLYNDGKEHKNFTFFC
KSGSSTETGSFESAAPFGWFLSTLPEPNQPIRLSHQGGAEITQFYFDKVKD

>Dromiciops_gliroides XP_043845630.1 interleukin-36 receptor antagonist protein
MVLQKALCFRMDAALKVIYVQNNQLLARRVQAGKPIDGEEISVVPNRLDAKRSPVILGLEGGTQCLSV
GIAQEPVLQIEHKNIMDLYRSKEESKSFTFYRMDTGLTSRLESAAYPGWFLCTIHEDEKPIITLNHPEDS
ESVITDFYFHQA

>Myotis_myotis XP_036167553.1 interleukin-36 receptor antagonist protein
MVLGALCFRMDAALKVLYLQDNQLLAGGVHAGKAVKGEEISVVPNRLDDSLSPVILGVQGSQCLSC
GMGQEPTLKLEPVDIMELYRSPKDSRGFTFYRRDTGLTSRFESAAPFGWFLCTVPEADQPLRLSQLPGDA
SWDHPIMDFYFQQCD

MSA (trimmed to remove overhangs):

CLUSTAL O(1.2.4) multiple sequence alignment

IL36G_HUMAN	EALEQGRGDPIYLGIQNPCLYCEKVGEQPTLQLKEQKIMDLYGGPEPVKPFIFYRAKT
Podarcis_cretensis	NRGMDEKNSPIFLGTQDGAQTLSCGESGGQPQLTLEHKAIMDLYNDGKEHKNFTFFCKSG
Trionyx_triunguis	GQWADGRSLPPSLQ---AVSPFLIGLS-HAPPPHPQDKEIMDLYQNSQEAKRFTFTCMAT
Pelodiscus_sinensis	NQFLDPSNFPIIMGIDNGTRCLSCGTS-APPTLMLEDKEIMDLYQNSQEAKRFTFTCVAT
Chelonia_mydas	NQFMDPSHFPIVMGIDGGLCCLSCSTS-AQPTLMLEDKKIMDLYQNSWEAERFTFTCSAS
Chelonoidis_abingdonii	NQFMDPSHFPIIMGIDGGTRCLSCGTS-AQPTLMLEDKKIMDLYQNSQEAKRFTFTCSAT
Emys_orbicularis	NQFMDPSHFPIIMGINGGTRCLSCGTS-AQPTLMLEDKKIMDLYQNSREAKRFTFTCSAT
Trachemys_scripta_elegans	NQFMDPSHFPIIMGINGGTRCLSCGTS-AQPTLMLEDKKIMDLYQNSQEAKRFTFTCSAT
Dromiciops_gliroides	NRLDAKRSPVILGLEGGTQCLSVGIA-QEPVLQIEHKNIMDLYRSKEESKSFTFYRMDT
Myotis_myotis	NRLDDSLSPVILGVQGSQCLSCGMG-QEPTLKLEPVDIMELYRSPKDSRGFTFYRRDT
	: * : : * : ** * . . * *
IL36G_HUMAN	GR--TSTLESVAFPDWFIASSK-RDQPIILTSELGKSY----NTAFELNIND-
Podarcis_cretensis	SSTETGSFESAAPFGWFLSTLPEPNQPIRLSHQGGA-----EITQFYFDKVKD
Trionyx_triunguis	GS--TRRFESVAFFPGWFLSTSPKDGQPLRVTNISGEA----EIINFYFKKM--
Pelodiscus_sinensis	GS--TRRFESVAFFPGWFLSTSPKDGQPVVTVTNISGEA----EIVNFYFKKM--
Chelonia_mydas	SS--TLRFELVAFPGWYLSTSPRNDQPLWVITYHLGEA----EITNFYFKKV--

```

Chelonoidis_abingdonii    GS--THRFEVAFPGWYLSTSPRNDQLQVTNRLGEA----EITNFYFKKV--
Emys_orbicularis         GS--TLRFESVAFPGWYLSTSPRNDQPLRVTNRLGEA----EITNFYFKKV--
Trachemys_scripta_elegans GS--TLRFESVAFPGWYLSTSPRNDQPLRVTNRLGEA----EITNFYFKKV--
Dromiciops_gliroides     GL--TSRLESAAYPGWFLCTIHEDEKPITLTNHPEDS--ESVITDFYFHQCA-
Myotis_myotis            GL--TSRFESAAPPGWFLCTVPEADQPLRLSQLPGDASWDHPIMDFYFQQCD-
.      *   :*  .:*.*:..:      :*  ::      *  :.

```

The multiple sequence alignment demonstrates that the novel *Trionyx triunguis* fragment clusters with IL36RN-like proteins across reptiles and mammals. Highly conserved motifs including IMDLYQNSQEAKRFTFTFC and FESVAFPGWFLSTSPK are shared with IL36RN orthologs but differ from IL36G, indicating that the novel sequence is most closely related to the IL-36 receptor antagonist subgroup of the IL-1 family.

Q6. Create a phylogenetic tree, using either a parsimony or distance-based approach. Bootstrapping and tree rooting are optional. Use “simple phylogeny” online from the EBI or any respected phylogeny program (such as MEGA, PAUP, or Phylip). Paste an image of your Cladogram or tree output in your report.

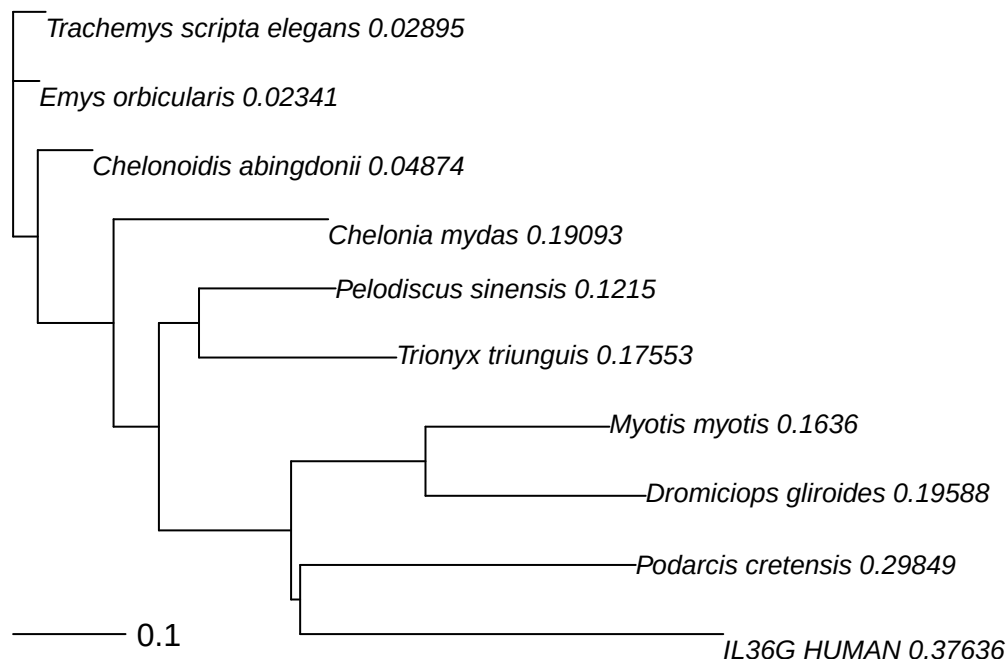


Figure 7: Phylogenetic Tree

Q7. Generate a sequence identity based heatmap of your aligned sequences using R. If necessary convert your sequence alignment to the ubiquitous FASTA format (Seaview can read in clustal format and “Save as” FASTA format for example). Read this FASTA format alignment into R with the help of functions in the Bio3D package. Calculate a sequence identity matrix (again using a function within the Bio3D package). Then generate a heatmap plot and add to your report. Do make sure your labels are visible and not cut at the figure margins.

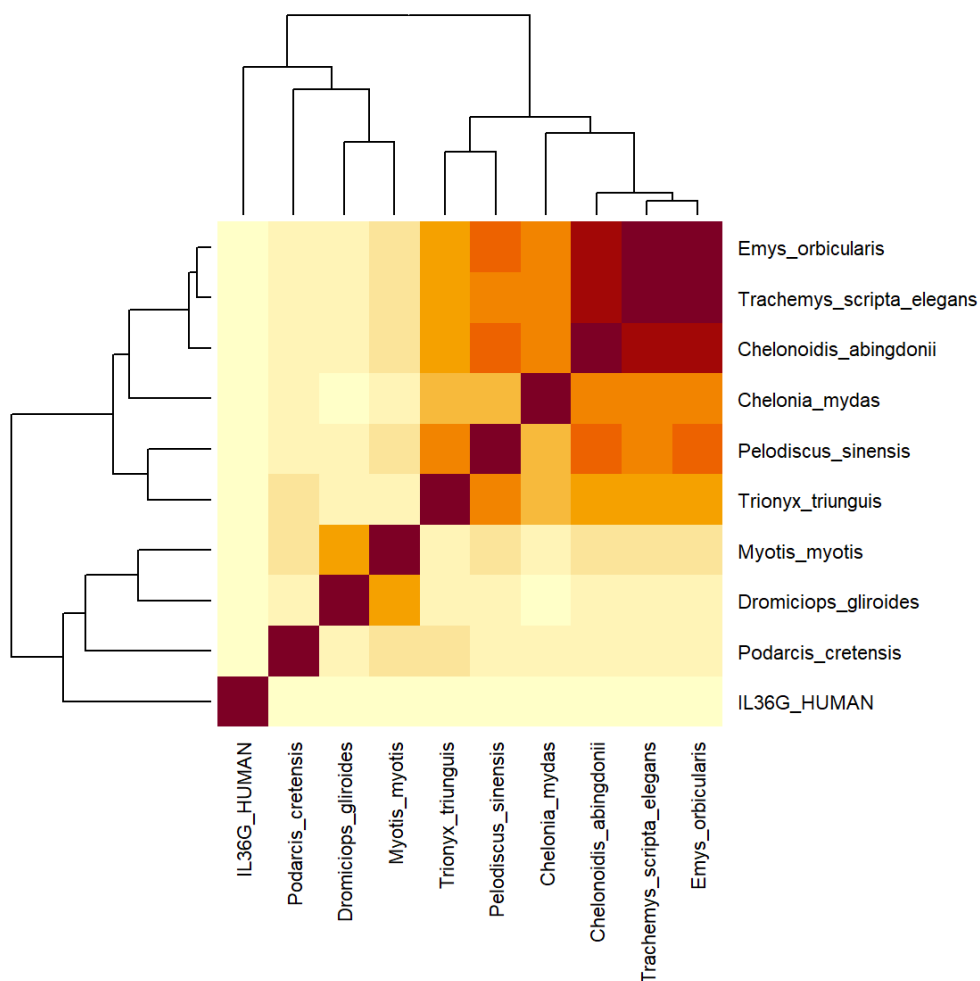


Figure 8: Clustered Heatmap

Q8. Using R/Bio3D (or an online blast server if you prefer), search the main protein structure database for the most similar atomic resolution structures to your aligned sequences.

List the top 3 unique hits (i.e. not hits representing different chains from the same structure) along with their Evalue and sequence identity to your query. Please also add annotation details of these structures. For example include the annotation terms PDB identifier (structureId), Method used to solve the structure (experimentalTechnique), resolution (resolution), and source organism (source).

HINT: You can use a single sequence from your alignment or generate a consensus sequence from your alignment using the Bio3D function `consensus()`. The Bio3D functions `blast.pdb()`, `plot.blast()` and `pdb.annotate()` are likely to be of most relevance for completing this task. Note that the results of `blast.pdb()` contain the hits PDB identifier (or `pdb.id`) as well as Evalue and identity. The results of `pdb.annotate()` contain the other annotation terms noted above. Note that if your consensus sequence

has lots of gap positions then it will be better to use an original sequence from the alignment for your search of the PDB. In this case you could chose the sequence with the highest identity to all others in your alignment by calculating the row-wise maximum from your sequence identity matrix.

	ID	Technique	Resolution	Source	Evalue	Identity
1	4P0J	X-RAY	2.30	Homo sapiens	2.12e-42	44.6
2	1MD6	X-RAY	1.60	Mus musculus	1.02e-38	44.1
3	4P0L	X-RAY	1.55	Homo sapiens	2.62e-38	41.8

Q9. Using AlphaFold notebook generate a structural model using the default parameters for your novel protein sequence. Note that this can take some time depending upon your sequence length. If your model is taking many hours to generate or your input sequence yields a “too many amino acids” (i.e. length) error you can focus on a single domain from your sequence - identify region by searching for PFAM domain matches.

Once complete save the resulting PDB format file for your records. Finally, generate a molecular figure of your generated PDB structure using the Mol* viewer online (or VMD/PyMol/Chimera if you prefer). To complete your analysis you should highlight conserved residues that are likely to be functional as spacefill and the protein as cartoon colored by local alpha fold pLDDT quality score. You can determine conserved residues from the alignment generated by the AlphaFold server and use a conservation cutoff appropriate for the diversity of your protein alignment (e.g. between 60% and 99% conserved). Note that pLDDT score is contained in the B-factor column of your PDB downloaded file. Please use a white or transparent background for your figure (i.e. not the default black in PyMol/VMD/Chimera etc.).



Figure 9: Spacefill Representation

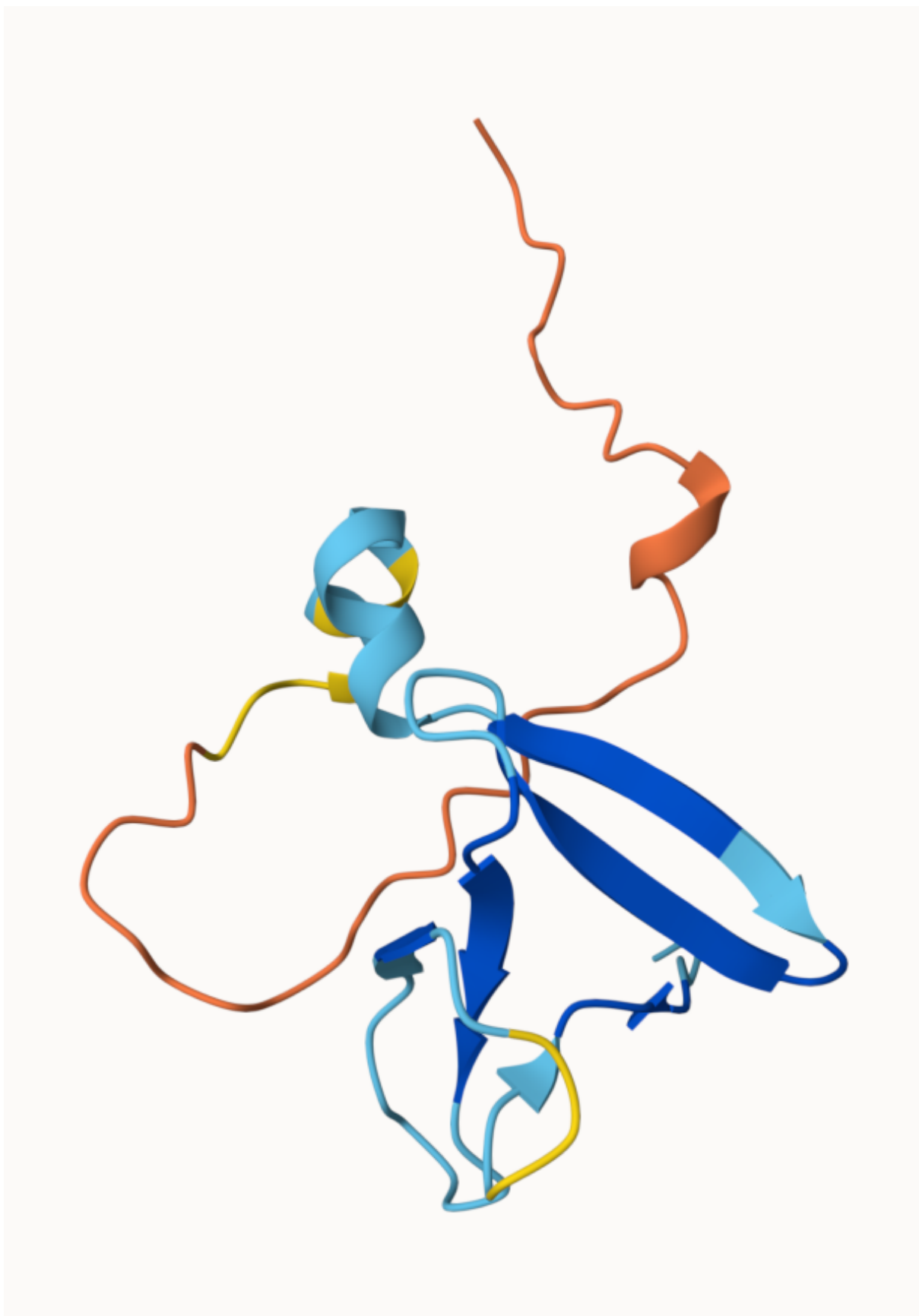
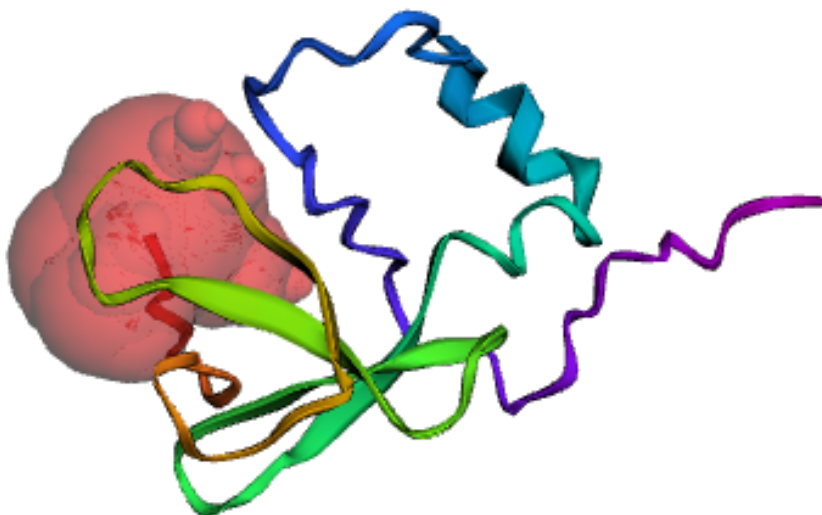


Figure 10: Base Representation

AlphaFold model shown as spacefill-highlighted and base cartoon representations.

Q10. (i) Using your computed structure model (or your closest homologue of known structure from the PDB) predict and locate potential small molecule binding sites using the CASTpFold server (<https://cfold.bme.uic.edu/castpfold/>). Provide an image or screen-shot of your largest predicted pockets “negative volume” and provide it’s area and volume.



The largest pocket by negative volume was Pocket 1, with a surface area of 255.621 Å² and a volume of 504.623 Å³.

- (ii) Perform a “Target” search of ChEMBL (<https://www.ebi.ac.uk/chembl/>) with your novel sequence. Are there any Target Associated Assays and ligand efficiency data reported that may be useful starting points for exploring potential inhibition of your novel protein? If there are no assays listed here simply list “non available as of [date]”.

Ans: After performing a biological sequence search with the novel protein sequence, several sequence-similar targets in the interleukin family were identified. The most relevant hit was interleukin-1 receptor antagonist protein from mouse (ChEMBL3414413; E-value 1.2e-11; 42.9% identity; 21 compounds, 23 activities). The ligand efficiency plot showed one data point with a Binding Efficiency Index (BEI) of 13.45 and a Surface Efficiency Index (SEI) of 4.78, with activity in the 1–100 nM range. This suggests that there are some known inhibitors for related proteins that could serve as starting points for exploring potential inhibition of the novel protein, although further experimental validation would be needed to confirm their relevance.

- (iii) Briefly discuss (100 words max) the druggability of your novel protein based on:

- Presence of well-defined pockets (output of tools like CASTpFold),
- Existence of known inhibitors for related proteins (your search of ChEMBL),
- Conservation of binding sites across homologs (your conservation analysis in Q10),
- Potential therapeutic applications if this protein were targeted (you can use ChatGPT, Claude etc. backed up by your reading of the literature here).

Ans: The novel protein appears to be moderately druggable. CASTpFold identified a well-defined pocket with a negative volume of 504.623 Å³, which is within the range of druggable sites. The ChEMBL search revealed known inhibitors with associated assay activity and ligand-efficiency data for related interleukin family proteins, suggesting that similar compounds could potentially target the novel protein. Conservation analysis showed that key residues in the predicted binding site are moderately conserved across homologs, indicating functional importance. Because the novel protein is most similar to an IL-36 receptor antagonist-like protein, targeting it would likely alter inflammatory signaling. Inhibiting it may increase inflammation by removing antagonist activity, whereas mimicking or enhancing it could help suppress excessive IL-36-mediated inflammation.